

Product and Category Management System

Project Title

Product and Category Management System using MySQL

1. Introduction

The Product and Category Management System is a simple database application used to store and manage product information in a shop or supermarket. The system allows users to organize products based on categories, maintain stock details, update product information, delete unwanted products, and generate reports category-wise.

This project is developed using SQL concepts such as table creation, primary key, foreign key, insert, update, delete, joins, and aggregate functions.

2. Objectives

- ❖ To create Category and Product tables.
 - ❖ To establish Primary Key and Foreign Key relationships.
 - ❖ To store product information.
 - ❖ To update and delete product records.
 - ❖ To generate category-wise product reports.
 - ❖ To understand relational database concepts.
-

3. Software Requirements

Software	Version
MySQL	8.0

MySQL Workbench	Latest
Windows	10/11

4. Database Name

```
CREATE DATABASE ProductManagement;  
USE ProductManagement;
```

5. Category Table

```
CREATE TABLE Category  
(  
  CategoryID INT PRIMARY KEY AUTO_INCREMENT,  
  CategoryName VARCHAR(100) NOT NULL,  
  Description VARCHAR(200)  
);
```

6. Product Table

```
CREATE TABLE Product ( ProductID INT PRIMARY KEY AUTO_INCREMENT,  
  ProductName VARCHAR(100) NOT NULL, CategoryID INT, Price DECIMAL(10,2), Stock  
  INT, FOREIGN KEY(CategoryID) REFERENCES Category(CategoryID) );
```

7. Insert Category Records

```
INSERT INTO Category(CategoryName,Description)
VALUES
('Electronics','Electronic Items'),
('Furniture','Home Furniture'),
('Stationery','Office Supplies'),
('Sports','Sports Equipment'),
('Groceries','Daily Needs');
```

8. Sample Output

Category Table

CategoryID	CategoryName	Description
1)	Electronics	Electronic Items
2)	Furniture	Home Furniture
3)	Stationery	Office Supplies
4)	Sports	Sports Equipment
5)	Groceries	Daily Needs

Product Table

ProductID	ProductName	Category	Price	stock
1)	Laptop	Electronics	65000	20
2)	Smart phone	Electronics	25000	35
3)	Office Chair	Furniture	4500	15

4)	Study Table	Furniture	7000	10
5)	Notebook	Stationery	80	200
6)	Pen	Stationery	20	500
7)	Cricket Bat	Sports	1800	25
8)	Football	Sports	1200	30
9)	Rice Bag	Groceries	1500	40
10)	Cooking Oil	Groceries	220	100

9. Advantages

- ❖ Easy to manage product details.
 - ❖ Reduces manual work.
 - ❖ Fast product searching.
 - ❖ Maintains stock information.
 - ❖ Category-wise reports can be generated.
 - ❖ Avoids duplicate data using relationships.
-

10. Applications

- ❖ Supermarket Management
 - ❖ Department Stores
 - ❖ Shopping Malls
 - ❖ Inventory Management
 - ❖ Retail Shops
 - ❖ Warehouse Management
-

11. Conclusion

The **Product and Category Management System** helps in managing product information efficiently using SQL. It demonstrates the use of relational database concepts such as primary keys, foreign keys, CRUD operations, joins, and aggregate functions. The system makes inventory management easier, improves data accuracy, and generates useful reports for better business decisions.